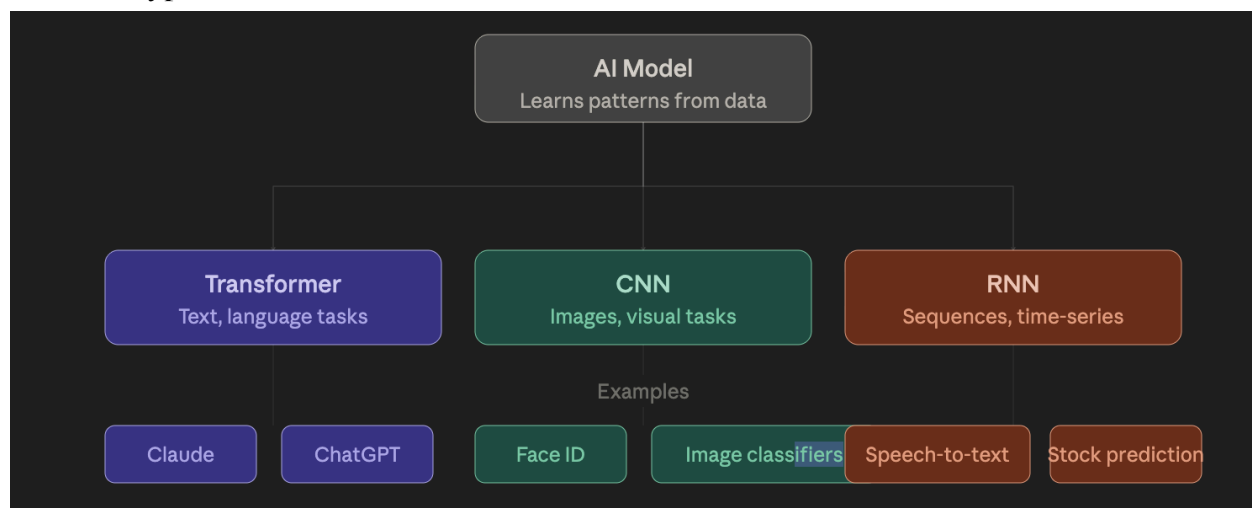


Training:

AI model reads, understands remembers the data - which is known as training, so when user asks(prompts) questions to AI model it answers(predicts), some times when we ask queries which were not part of the training data then AI model generates raw facts and may does incorrect answers(predictions) to user questions this is known as **hallucination**.

- **For ex.** student studies for exams using book A. In the exam, if the student encounters the questions which he read from book A he will answer(predict) it correctly, but if he encounters a question which was not present in book A during studying then he tries to predict answers which might be correct or wrong.

AI model types:



AI Model: It is a software program that is trained on (reads) data to recognize patterns and make decisions(answers/predictions). AI model only understands data in number format (known as token ID) **Example:** The **algorithm** that recognises your face to unlock your phone.

LLM / Large Language Model: Since AI model can understand only data in number format, humans cannot query (prompt) ai model in human readable text format, User enters prompt to LLM, LLM internally has tokenizer that uses vocab and generates tokens ids from the human prompt and sends the generated token IDs (number data) to transformer i.e a type of AI model. the response from the transformer will be in number format, transformer sends the response in number data format to tokenizer, tokenizer converts it back to human readable format and sends the human readable response back to user through LLM. **Example:** Every LLM like ChatGPT, Claude has tokeniser to generate number data from human readable prompt and vice versa.

Transformer: It is a component present in the LLM, it is a type of AI model, it is used to do text based tasks like human readable query prompts.

Tokenizer:

The human prompt is given to LLM,

The tokenizer is an internal component in LLM.

Tokenizer is an algorithm/ concept used to split the human prompt into smaller chunks which is known as tokens

The ML Engineer at the AI Research Organisation (like Google, OpenAI, Meta) trains the Tokenizer — the same team that builds the LLM itself.

The ML Engineer collects the raw dataset — billions of text documents (Wikipedia, books, GitHub code, websites).

They run a Tokenizer training algorithm (most commonly Byte Pair Encoding / BPE/..) on that raw text. This is one of the algorithm that chatgpt uses. This algorithm is used to generate tokenIds for the prompt using vocab.

Vocabulary:

Think of the Vocabulary as a giant dictionary that the AI memorized.

It doesn't contain meanings; it only maps **text chunks** to **ID numbers**.

It has a fixed size: Most AI vocabularies have exactly 100,000 slots. It cannot learn new words once it is built.

It contains whole words: The most common words in human language get their own slot.

Slot 104 = "The"

Slot 8241 = "Who"

It contains parts of words: Because there are way more than 100,000 words in the world, the AI uses the leftover slots to store syllables, prefixes, and letters so it can build words that didn't fit.

Slot 502 = "ing"

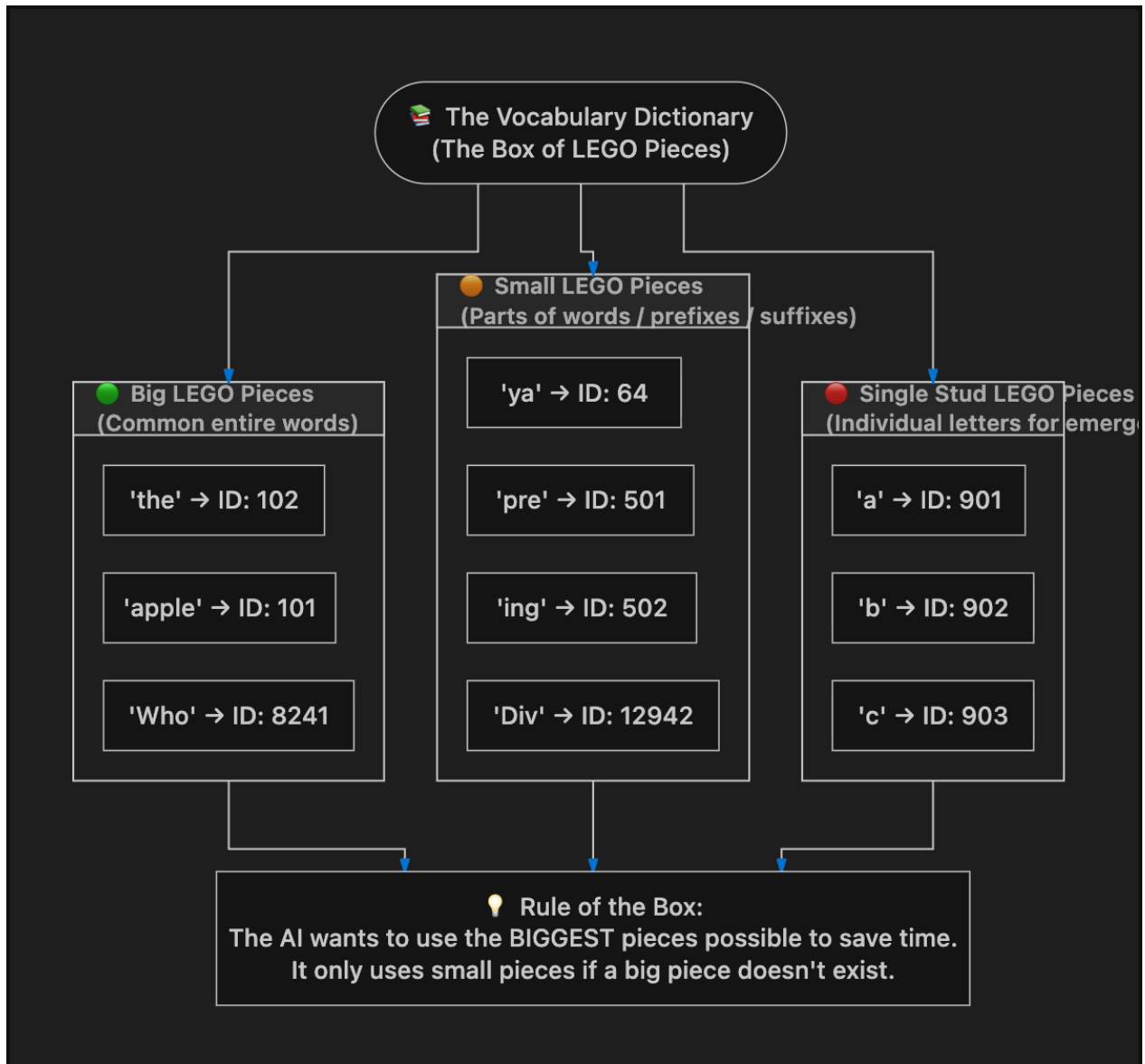
Slot 12942 = "Div"

Slot 64 = "ya"

it has slots for all punctuation marks and english characters,..

slots are nothing but token ids

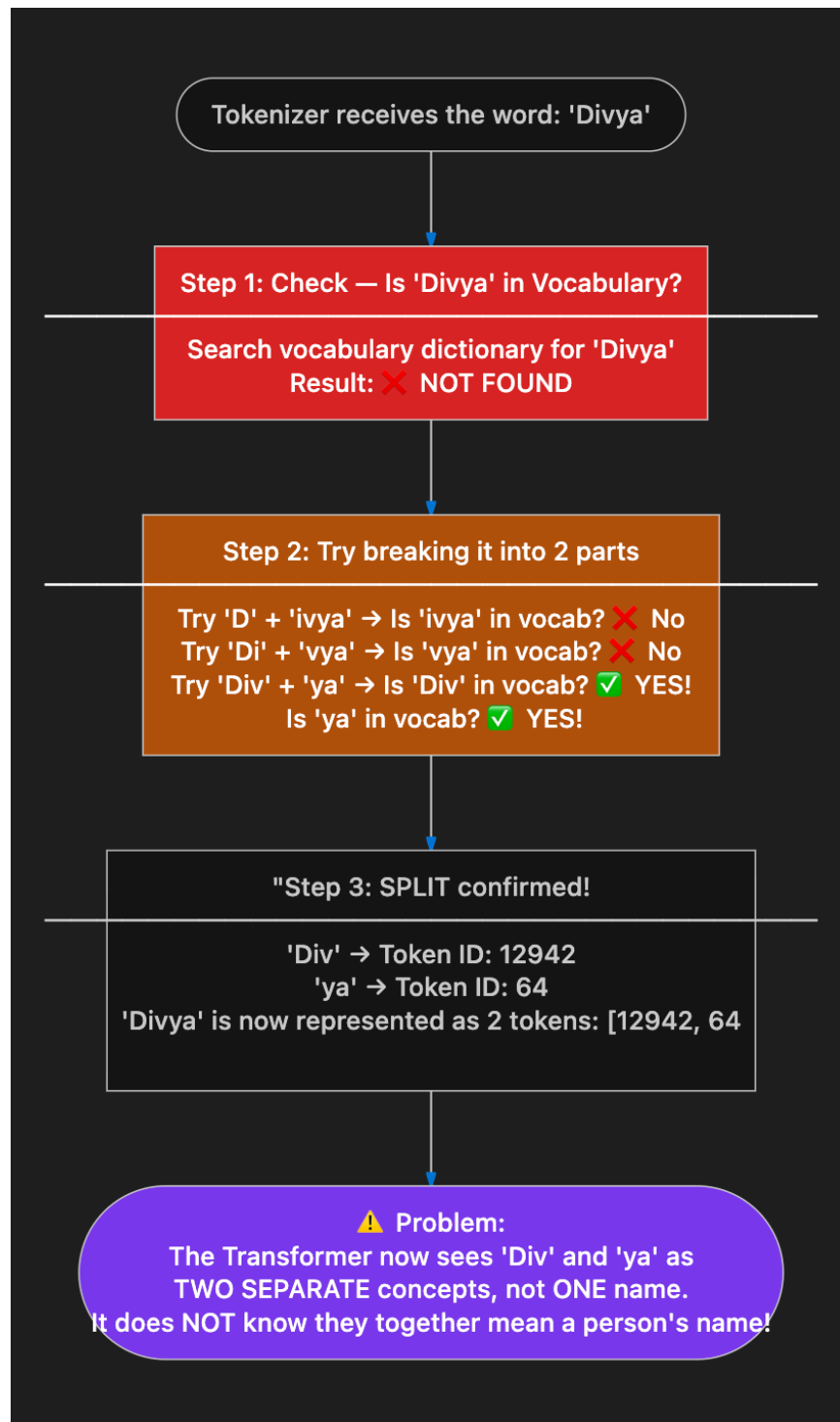
Vocab content:



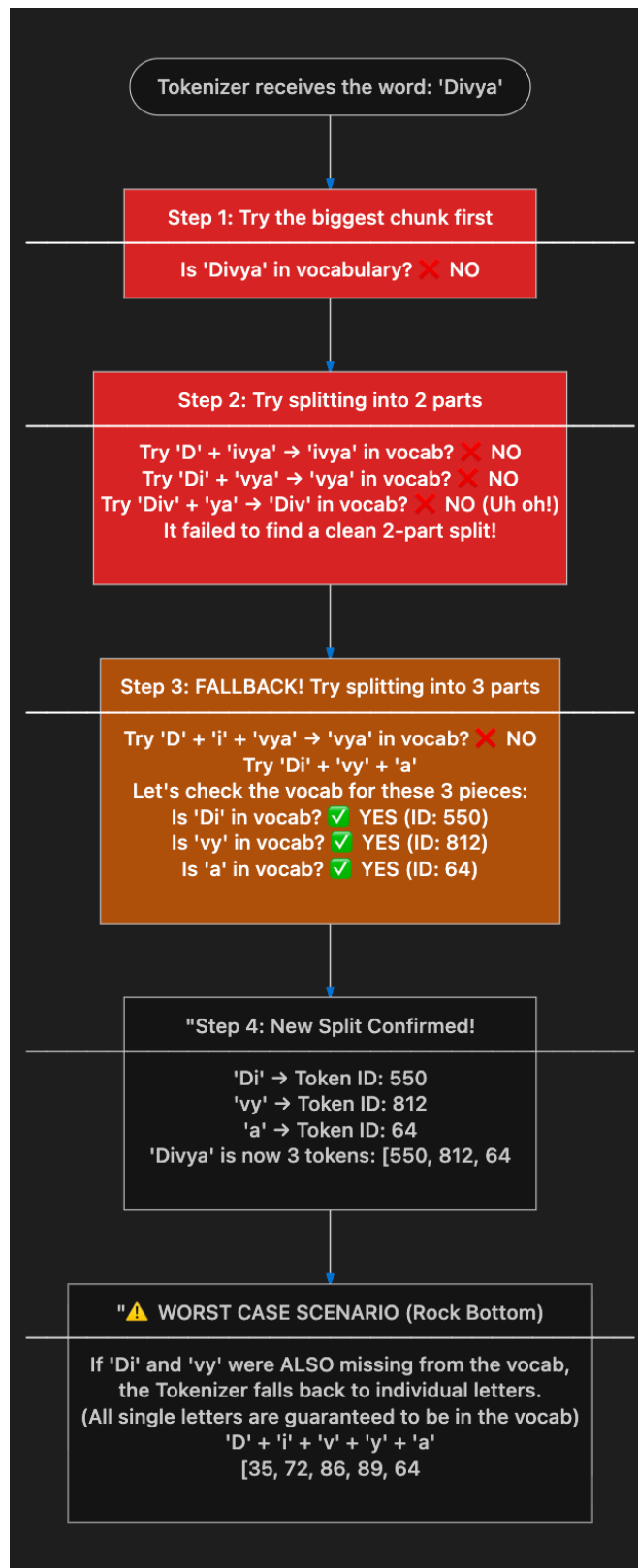
Tokenizer working:



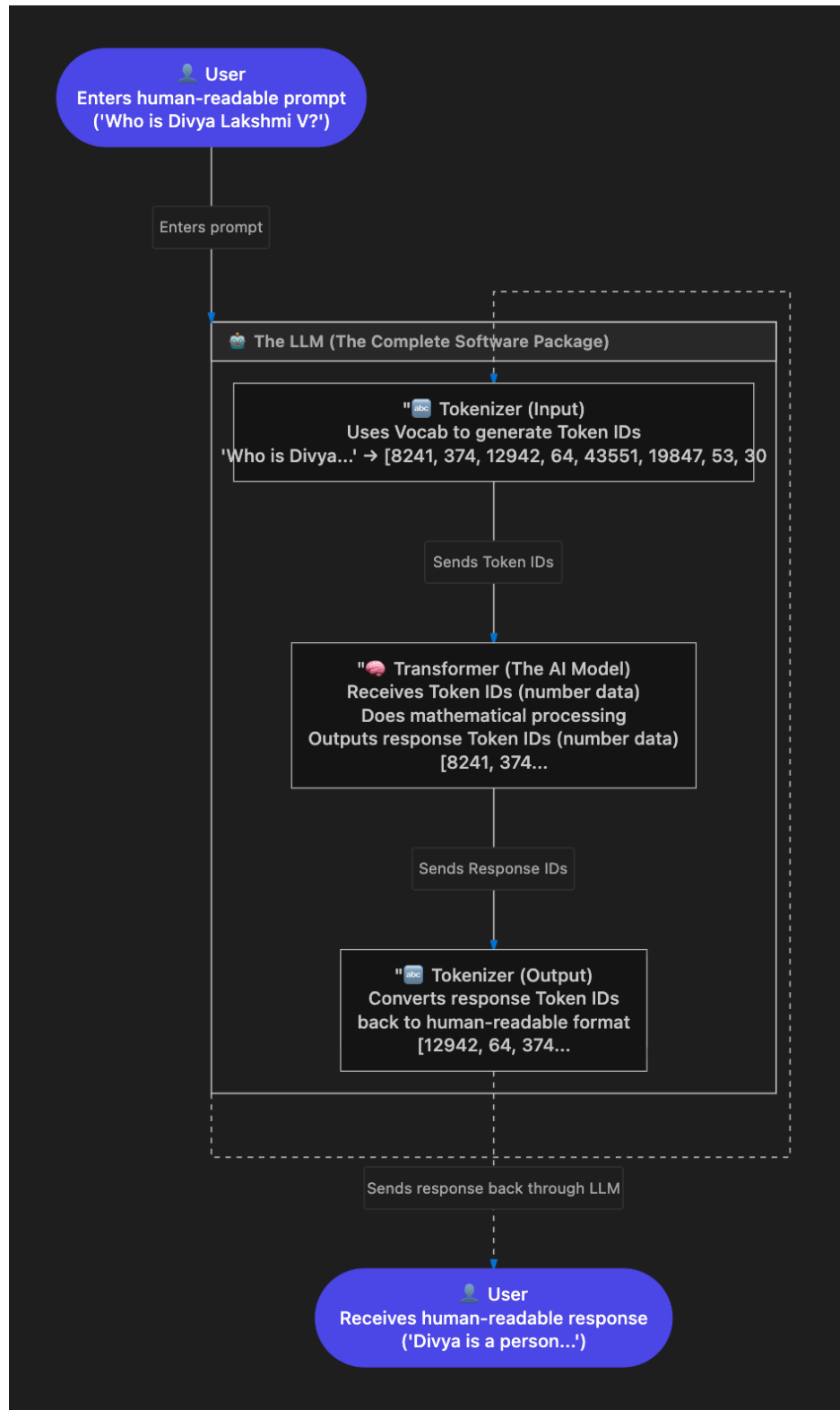
How Divya" is Split Step-by-Step:



What if "Div" is NOT in the Vocabulary



Working of LLM:



Context window: (Notes on TokenIds):

if more token ids are sent to transformer/ai model, the response time also will be more.
the less the token ids sent to transformers/ ai model the less the response time would be.
for ex.

chatgpt has 128K limit token ids for both request and response together.
128k means it means at a time it can read and process 300-400 pages of a book
if you exceed this limit the chatgpt will tell token limit exceed like something.

Note on LLM:

when you use copilot in office, sometimes when you pasted more content in the prompt, you can see "sorry something went wrong" error, that is because of company's fire wall has restricted that copy pasting more lines in prompt. the no. of lines /words can pasted/ entered in copilot prompt can be restricted/ limited by a firewall, so that user cannot give more company data/code at a time in a prompt